AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I051*

Team Name: *YUKTHI*

Domain: *Domain 1 – Intelligent Systems & Digital Futures*

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Problem Statement

Migrant workers often lose access to government benefits after moving to another state because they do not know which schemes they qualify for, what documents they need, or how to apply in the new state, and because language barriers make official information hard to understand.

Problem Understanding

- **Who:** Internal migrant workers (construction, factories, domestic work, delivery, agriculture) and their families, including children.
- **Causes:** Schemes are run by separate departments and portals; official information is text-heavy and not in native languages; rules and documents change from state to state.
- **Effect:** Eligible families miss scholarships, health insurance, pensions and accident cover. Missing documents or missed deadlines cancel benefits, and workers depend on agents or middlemen for incomplete advice.

Existing Gaps & Challenges

- Government portals (including e-Shram and One-Stop-Solution) are text-heavy and English-focused; workers may not know how to search, check eligibility or complete applications.
- Helplines, offices, NGOs and unions are limited by working hours, wait times, language, travel, staff and scale; employers, agents, friends and family give incomplete or outdated advice.
- **Unsolved:** No single, simple, multilingual system helps workers discover eligible schemes, understand documents and apply with confidence.

Our Proposed Solution

- **Core idea:** A voice-first, multilingual scheme navigator that workers use by voice or WhatsApp in their own language.
- **How it works:** A rules engine over 10–15 high-impact schemes (sources shown) returns eligible schemes, a missing-document checklist and where to get each proof; a verified helper can be reached with a trackable case ID.
- **Value:** Workers move from eligible on paper to actually claiming benefits, without depending on middlemen.

Technical Approach

Worker (voice / WhatsApp / SMS) → **Speech layer** (Bhashini / IndicASR) → **Rules engine + RAG** over scheme documents → **Output:** eligible schemes, missing-document checklist, helper with case ID

The worker speaks in their own language; speech is converted to text and matched against scheme rules, and the answer returns as voice or text.

Tech Stack

Frontend: Flutter / React Native

Backend: FastAPI

Database: PostgreSQL, vector database

AI/ML, APIs, or Hardware: Bhashini / IndicASR, RAG, WhatsApp / SMS gateway

Innovation & Uniqueness

- **New:** One voice conversation takes a worker from "which scheme applies?" to a personalised document checklist and a human helper, instead of leaving them to search portals.
- **Why it fits:** Voice, WhatsApp and SMS channels in the worker's own language suit low digital literacy and remove the language barrier.
- **Differentiator:** A rules engine with sources shown, so every answer traces to official scheme documents, plus a verified helper with a trackable case ID.

Feasibility & Scalability

- **Feasibility:** Software-only; needs laptops, a cloud-hosted server, Android phones for voice testing, public scheme documents and Bhashini API access. No specialised hardware.
- **Scalability:** Pilot one worker group in one state corridor with 10–15 schemes; grow by adding schemes, states and languages as data and rules.
- **Risks:** Scheme rules vary by state and change; mitigated by showing sources and curating data with NGO, labour department and union mentors.

Cost for Product (Optional)

- Development or hardware cost: [Add estimate]
- Hosting or maintenance cost: [Add estimate]
- Estimated total: [Add amount and currency]

Use Cases (compulsorily 3 use cases must be there)

- Use case 1: A construction worker who has moved to a new state asks, by voice in their own language, what they qualify for, and gets eligible schemes plus a missing-document checklist.
- Use case 2: A migrant family with school-age children finds scholarship and school admission support and learns which proofs to carry and where to get them.
- Use case 3: A domestic worker stuck on an application reaches a verified helper (CSC, NGO or union) and tracks progress with a case ID.
  Note: Additional slides may be included only for this section.

Impacts and Benefits

- **Workers:** Find eligible schemes in one voice chat, know exactly which documents are missing, and avoid repeat office visits and middlemen.
- **Society:** Scholarships, health insurance, pensions and accident cover reach the families they are meant for, reducing financial stress in illness or job loss.
- **Measured by:** Time to complete an eligibility check, share of users with a full document checklist, and applications started and benefits received.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 1	Target 1.3	No Poverty	Connects workers to social protection schemes
SDG 8	Target 8.8	Decent Work and Economic Growth	Protects migrant workers' rights and entitlements
SDG 10	Target 10.7	Reduced Inequalities	Supports safe, well-managed migration

Individual Member Contribution

Member	Contribution or responsibiliTY
Bhalasubramaniyan R	[TEAM LEADER]
Niranjan S	[TEAM MEMBER 1]
Shagarika P	[TEAM MEMBER 2]
Pooja S	[TEAM MEMBER 3]

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